Did the Phillips Curve "Fail" during the Pandemic? Expectations, Slack and OOS Evidence

1. Abstract

This note asks whether the Phillips curve "failed" to predict U.S. inflation after COVID, or whether standard implementations omitted key post-2020 forces. Using quarterly U.S. data (1968Q1-2023Q3), I estimate expectations-augmented Phillips curves on a strict pre-2020 training sample and evaluate forecasts over 2020Q1-2023Q3 against inflation-only benchmarks. Post-2020 tracking is highly sensitive to the expectations proxy: Michigan household expectations outperform SPF and market-based measures. Baseline models do not reliably beat persistence benchmarks, but forecast accuracy improves materially when slack is remeasured using vacancy tightness and cost-push proxies (energy/import prices) are included.

2. Introduction

The post-pandemic inflation surge reopened a basic question in macroeconomics: did the Phillips curve "fail" to predict inflation after COVID, or did standard implementations omit forces that became first-order in 2020-23? In modern New Keynesian models, inflation depends on expected inflation and real marginal costs, up to a cost-push disturbance (Clarida, Galí, & Gertler, 1999). Read as a reduced-form forecasting relationship, the post-2020 episode appeared to represent a breakdown, motivating disciplined pseudo out-of-sample evaluation (Stock & Watson, 2010).

This note evaluates the post-pandemic "failure" claim using quarterly U.S. data from 1968Q1–2023Q3 and a strict split (train through 2019Q4; test 2020Q1–2023Q3) for headline PCE inflation (with core PCE as robustness). Two recent contributions provide the central motivation. Beaudry, Hou, and Portier (2025) argue that post-pandemic inflation can be reconciled with a Phillips-curve framework once short-horizon expectations and shocks are jointly accounted for, using Michigan household expectations as a baseline proxy. Hazell (2025), in contrast, stresses that post-2020 conclusions are highly sensitive to the expectations measure itself: swapping household, professional, and market-based proxies can produce sharply different inferences about "Phillips-curve failure." I therefore begin with an expectations "horse race," holding the slack proxy fixed while varying expectations measures, and evaluate models strictly out-of-sample.

To make "failure" operational, I benchmark Phillips-curve forecasts against simple inflation-only persistence models (no-change and AR-type forecasts), following the forecasting literature's warning that Phillips curves often add limited incremental predictive content (Atkeson & Ohanian, 2001). I then test whether forecast performance improves when slack is remeasured using vacancy-based tightness and when parsimonious supply/relative-price controls are added, reflecting prominent pandemic-era accounts emphasising bottlenecks and energy/import-price movements (Di Giovanni et al., 2022; Bernanke & Blanchard, 2023) and evidence that vacancy tightness can be more informative than the unemployment gap (Barnichon & Shapiro, 2024). As a compact diagnostic, I also test whether

a simple quarterly fiscal impulse proxy adds incremental predictive content once expectations, tightness, and supply controls are included.

Previewing the results, post-2020 tracking is highly sensitive to expectations measurement, the baseline expectations-plus-unemployment-gap Phillips curve underpredicts the 2021-22 surge, and forecast performance improves materially when vacancy tightness and supply proxies are incorporated; by contrast, our fiscal extension does not deliver robust out-of-sample gains and often introduces systematic bias.

3. Data

The analysis uses a combined quarterly US macroeconomic dataset covering 1968Q1-2023Q3, compiled from standard macroeconomic sources, including the BEA National Income and Product Accounts and FRED as a distributor (U.S. Bureau of Economic Analysis, 2026; Federal Reserve Bank of St. Louis, 2026). The core inflation and labour-market series span the full sample, while some expectations measures start at a later date. As a result, specifications that rely on different expectations proxies are estimated on different effective samples.

The main outcomes are year-on-year headline PCE inflation and, for robustness, core PCE inflation, constructed from the PCE price indices in the dataset (*pceall and PCEexcludingfoodandenergy*). In the baseline Phillips-curve specifications, inflation is related to a proxy for expected inflation, drawn from household expectations in the Michigan Survey (Surveys of Consumers, University of Michigan, 2026), professional forecasts from the Survey of Professional Forecasters (Federal Reserve Bank of Philadelphia, 2026), and, where available, market-based inflation swaps (Bloomberg L.P., 2026). Inflation is also related to a proxy for slack, measured either by an unemployment-gap concept based on unemployment relative to a natural-rate estimate or by vacancy-based labour-market tightness using the vacancy-unemployment ratio ($log(V/U)$). To capture cost-push and relative-price forces that may be especially important for headline inflation after 2020, I additionally use supply-side proxies in the dataset, including PCE energy prices, the import deflator (U.S. Bureau of Economic Analysis, 2026), and Brent oil prices (International Monetary Fund, 2026), from which inflation rates are constructed where required. For the fiscal extension, I augment the dataset with quarterly federal current expenditures and receipts and nominal GDP from the BEA National Income and Product Accounts via FRED to construct a deficit-to-GDP measure and a fiscal impulse proxy aligned to the year-on-year inflation horizon (Federal Reserve Bank of St. Louis, 2026).

4. Empirical Strategy

4.1 Baseline Phillips curve and Expectations measurement

I begin with the expectations-augmented Phillips curve used as a reduced-form forecasting relationship:

$$\pi_t = \alpha + \beta\pi_t^e - k\widehat{u}_t + \varepsilon_t \quad (1)$$

Where π_t is year-on-year PCE inflation, π_t^e is a one-year-ahead inflation expectation, and $\widehat{u_t}$ is an unemployment gap measure. In the baseline, slack is defined as unemployment relative to a short-term natural rate estimate:

$$\widehat{u_t} = u_t - u_t^*, \text{ where } u_t = UNRATE_t, u_t^* = NROUST_t$$

A central point of the post-pandemic Phillips-curve debate is that π_t^e is not directly observed, and empirical conclusions can depend on which expectations proxy is used. I therefore estimate (1) while holding the slack measure fixed and varying only π_t^e across the one-year measures available in the dataset: Michigan household expectations (mean and median), professional forecasts from the SPF (one-year-ahead), and the one-year inflation swap rate (available only from 2004).

This subsection isolates how much the baseline Phillips curve's post 2020 performance depends on how expected inflation is measured. To assess whether any baseline specification is genuinely useful in the sense of the research question, the next subsection evaluates predictive content out of sample and relative to benchmark forecasts.

4.2 Benchmarking Out-of-sample tests

To interpret "failed to predict" literally, I evaluate Phillips-curve specifications in a strict pseudo out-of-sample framework and compare them to simple inflation-only benchmarks, following the inflation forecasting literature (Atkeson & Ohanian, 2001; Stock & Watson, 2008). Models are estimated on data through 2019Q4 and evaluated over 2020Q1–2023Q3. For each specification *m,* forecast errors are:

$$e_t^{(m)} = \pi_t - \hat{\pi}_t^{(m)} \tag{2}$$

I implement three OOS diagnostics. First, I run a benchmark contest in the COVID window, comparing Phillips-curve forecasts to standard univariate alternatives: a no-change forecast $\hat{\pi}_t^{NC} = \pi_{t-1}$, a trailing mean $\hat{\pi}_t^{MA} = \frac{1}{K}\sum_{j=1}^{K} \pi_{t-j}$, and an AR(1) forecast $\hat{\pi}_t^{AR1} = \hat{c} + \hat{\rho}\pi_{t-1}$ estimated on the training sample. Second, I examine window sensitivity by repeating the same evaluation across alternative test periods (post-GFC versus post-COVID versus a longer post-2008 window), since the forecasting literature emphasises regime dependence.

These benchmark and robustness checks provide a disciplined standard for "failure": if baseline Phillips-curve forecasts are close to benchmark performance, the interpretation is limited to incremental predictive content beyond inflation persistence. The next subsection then treats any remaining systematic post-2020 errors as a specification and identification problem, asking whether alternative slack measures and cost-push proxies improve OOS performance.

4.3 Specification and Identification extensions

The benchmark comparisons in Subsection 4.2 imply that baseline Phillips-curve regressors often add limited incremental forecasting power beyond inflation persistence, consistent with the broader inflation forecasting literature. A constructive interpretation is therefore an identification/specification one: post-2020 forecast errors may be driven by omitted shocks and measurement issues that a canonical unemployment-gap Phillips curve does not capture.

4.3A Cost-push augmentation: Hazell-style energy and bottlenecks-style import prices

I first allow for an observable cost-push component, augmenting the baseline expectations-augmented Phillips curve with a supply proxy z_t:

$$\pi_t = \alpha + \beta\pi_t^e + \kappa \cdot slack_t + \gamma z_t + \varepsilon_t \tag{3}$$

Following Hazell's post-pandemic interpretation (Hazell, 2025), the primary "Hazell-style" control is PCE energy inflation ($z_t = \pi_t^{energy}$). I then use import price inflation ($z_t = \pi_t^{import}$) as a "bottlenecks/supply-chain" proxy (Di Giovanni et al., 2022). The bottlenecks view stresses that, when global production and shipping capacity become temporarily constrained, firms face sharp increases in the marginal cost of traded intermediate inputs and imported final goods. Therefore, these variants test the core identification claim: whether the post-2020 "miss" arises because the baseline Phillips curve omits large cost-push/relative-price shocks that shift the inflation process.

4.3B Slack remeasurement: unemployment-gap slack versus labour-market tightness

A second identification concern is that the unemployment gap may be an incomplete proxy for marginal cost in the post-pandemic labour market. When matching efficiency changes or the Beveridge curve shifts, unemployment can remain relatively low even as firms face acute hiring difficulty, upward wage pressure, and faster price adjustment. Vacancy-based tightness, commonly measured by the vacancy-unemployment ratio V/U, is therefore often argued to be a more relevant indicator of inflationary pressure than unemployment alone in such episodes (Barnichon & Shapiro, 2024). I therefore replace unemployment-gap slack with labour-market tightness and estimate:

$$\pi_t = \alpha + \beta\pi_t^e + \lambda \cdot log(V/U)_t + \varepsilon_t \tag{4}$$

And then re-estimate the same tightness specification with the cost-push controls from before.

I use $log(V/U)_t$ rather than the level of V/U as a functional-form choice. Taking logs makes the tightness measure more comparable to a 'gap' variable, reduces the influence of extreme post-2020 values, and yields coefficients that can be interpreted as semi-elasticities, which is convenient for comparing magnitudes across specifications.

4.3C Robustness

I then use targeted robustness checks that speak to interpretation rather than adding new model families: (i) expectations-regime dependence (Michigan versus SPF) in cross-era OOS validation, (ii)

repeating key specifications for core PCE inflation to verify that "energy rescues" are specific to headline inflation rather than mechanically fitting energy components.

Across all variants and robustness checks, the evaluation criterion remains consistent: improvements must show up as lower post-2020 OOS RMSE and reduced systematic bias, not only stronger in-sample fit.

4.4 Fiscal extension: a fiscal-impulse proxy and FTPL motivation

A further hypothesis is that pandemic-era inflation partly reflected fiscal expansion. FTPL-type mechanisms imply that if future primary surpluses are perceived as insufficient, higher expected deficits can be absorbed by a higher price level (Leeper, 1991). I do not attempt to identify fiscal 'news' shocks; instead, I run a diagnostic: conditional on expectations, tightness, and supply proxies, does a simple fiscal stance measure add incremental predictive content for post-2020 inflation?

I therefore augment the preferred Phillips-curve specification with a fiscal impulse proxy. Let $d_t = (G_t - T_t)/Y_t$ denote the deficit share, where G_t is federal current expenditures, T_t federal current receipts, and Y_t nominal GDP. I use the four-quarter change $\Delta_4 d_t$ to match the year-on-year inflation horizon and to capture persistent fiscal expansions rather than short-lived quarterly timing effects. I then estimate the pre-2020 model and evaluate out-of-sample forecasts over the post-2021 window where $\Delta_4 d_t$ is observed:

$$\pi_t = \alpha + \beta\pi_t^e + \lambda log(V/U)_t + \gamma z_t + \delta\Delta_4 d_t + \varepsilon_t$$

with z_t denoting the baseline cost-push proxy (energy inflation, or import inflation in the corresponding variant).

This extension is interpreted as predictive content, not causality: the deficit impulse $\Delta_4 d_t$ is endogenous to the same shocks driving inflation and labour-market conditions. For example, a demand boom can raise inflation and tighten the labour market while also improving tax receipts, mechanically reducing the deficit share.

5. Results and Discussion

5.1 Expectations horse-race

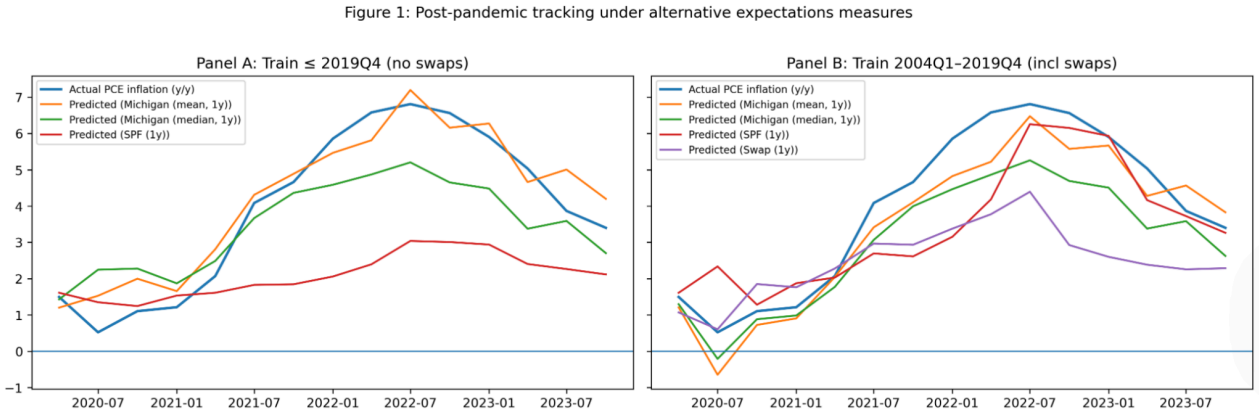

Figure 1: Baseline NKPC tracking plot with different expectations measures (Panel 2 trains on a shorter training set as we include market-based swaps which only has data available from 2004)

Expectations measurement is first-order for post-pandemic tracking. Figure 1 compares baseline expectations-augmented Phillips-curve forecasts that differ only in the one-year expectations proxy. In Panel A (train ≤2019Q4), Michigan household expectations deliver substantially better post-2020 tracking than SPF 1-year expectations, which remains far below realised inflation during the 2021-22 surge. Panel B repeats the exercise on a common 2004Q1-2019Q4 training window and adds the 1-year inflation swap; the swap-based forecast underpredicts even more sharply. The key point is diagnostic: the perceived "Phillips curve failure" is highly sensitive to how expected inflation is measured, consistent with Hazell's emphasis that Phillips-curve conclusions can be knife-edge in expectations measurement and with Beaudry-Hou-Portier's focus on the joint movement of expectations and shocks.

5.2 OOS test forecasting

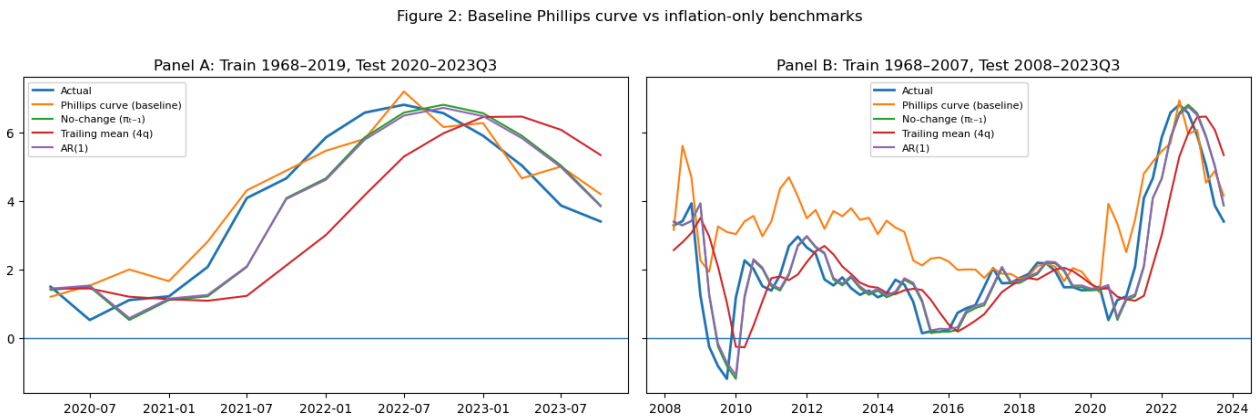

Figure 2: Phillips curve baseline (Michigan mean expectations) vs inflation persistence benchmarks, Panel A (train set between 1968-2019), Panel B (train set between 1968-2007)

Baseline Phillips-curve forecasts do not clearly dominate persistence benchmarks, and conclusions are window-dependent. Figure 2 evaluates the baseline Phillips curve against inflation-only benchmarks in a tracking format. In the COVID window (Panel A), no-change, trailing-mean and AR(1) benchmarks track inflation surprisingly well through much of the episode, and the baseline Phillips curve does not robustly outperform them at the turning points. Extending the test window (Panel B) makes regime dependence explicit: the baseline specification behaves very differently across the low-inflation post-GFC era and the post-pandemic surge. This aligns with the inflation forecasting literature's message that Phillips-curve forecasts often struggle to beat simple univariate benchmarks in pseudo out-of-sample exercises (Atkeson & Ohanian, 2001; Stock & Watson, 2008), so "failure" should be interpreted as limited incremental predictive content beyond inflation persistence rather than a definitive rejection of the mechanism.

5.3: Model Specification

Table 1: Phillips curve variants (OOS 2020Q1–2023Q3)

	Baseline (exp + u-gap)	+ Energy (Hazell)	+ Import (bottlenecks)
Panel A: Unemployment-gap Phillips curves			
Slack slope	-0.002	0.035	0.080
	(0.105)	(0.111)	(0.126)
RMSE	0.633	0.618	0.708
Bias	0.267	0.373	0.111
Train N	204	204	204
Test N	15	15	15
	Tightness (exp + log V/U)	Tightness + Energy	Tightness + Import
Panel B: Tightness Phillips curves (log V/U)			
Slack slope	-0.041	-0.082	-0.183
	(0.318)	(0.316)	(0.343)
RMSE	0.636	0.563	0.625
Bias	0.238	0.305	-0.031
Train N	204	204	204
Test N	15	15	15

Figure 3/Table 1: Reports accuracy metrics and coefficients for different Phillips curve variants (after varying slack and supply chain/bottleneck controls), "Bias" here refers to the mean forecast error, where a value > 0 indicates overprediction on average and vice versa. This is consistent across the paper.

Two extensions account for much of the post-2020 miss: cost-push controls and tightness-based slack. Table 1 treats the baseline underprediction as a specification problem. In the unemployment-gap Phillips curve, the baseline (expectations + u-gap) underpredicts headline inflation (RMSE 0.633; bias 0.267). Adding supply proxies improves performance, with import-price inflation in particular reducing bias more than energy in the unemployment-gap formulation. A natural interpretation, consistent with bottleneck accounts, is that import prices proxy broader tradables and supply-chain pressures that propagate beyond energy components and were important early in the inflation surge (di Giovanni et al., 2023).

The more decisive change is slack measurement. Replacing the unemployment gap with vacancy tightness $log(V/U)$ yields stronger post-pandemic tracking once cost-push forces are controlled for: tightness + energy delivers the best headline accuracy in this table (RMSE 0.563), while tightness + import produces near-zero bias (-0.031). This is intuitive if the Beveridge curve shifts or matching

efficiency changes: unemployment gaps can understate inflationary pressure when vacancies remain unusually elevated, whereas V/U more directly captures labour scarcity and associated wage/marginal-cost pressure (Barnichon & Shapiro, 2024).

One additional diagnostic consistent with Hazell's emphasis on expectations measurement is that Michigan mean tracks the surge more closely than the median, implying a smaller incremental 'rescue' from supply controls when the mean is used as π_t^e. *(See Appendix Figure 1)*

However, a post-2020 'rescue' is only persuasive if it behaves sensibly across regimes and across inflation concepts, so the next exhibit tests regime dependence (Michigan versus SPF) and contrasts headline with core.

Table 2: Robustness (regime dependence and inflation target)

	RMSE	Bias
Panel A: Regime dependence (same model, different expectations proxy)		
Model: $\pi_t \sim \pi_t^e + \log(V/U)_t + \pi_t^{energy}$		
Test 2008–2019 (train 1968–2007) Michigan mean	1.574	1.223
Test 2008–2019 (train 1968–2007) SPF 1y	0.357	-0.072
Test 2020–2023Q3 (train 1968–2019) Michigan mean	0.563	0.305
Test 2020–2023Q3 (train 1968–2019) SPF 1y	1.905	-1.620
Panel B: Inflation target (headline vs core)		
Headline PCE (y/y): $\pi_t \sim \pi_t^e + \log(V/U)_t + \pi_t^{energy}$ OOS 2020–2023Q3	0.563	0.305
Core PCE (y/y): $\pi_t \sim \pi_t^e + \log(V/U)_t + \pi_t^{energy}$ OOS 2020–2023Q3	0.652	0.131
Core PCE (y/y): $\pi_t \sim \pi_t^e + \log(V/U)_t + \pi_t^{import}$ OOS 2020–2023Q3	0.593	0.193

Figure 4/Table 2: Reports accuracy metrics for two different robustness tests (Varying expectations with best performing PC variant and varying inflation target)

Table 2 shows the post-pandemic "best" specification is not regime-invariant, which matters for interpreting "failure." Holding the model structure fixed, Michigan-based expectations produce large overprediction in the post-GFC low-inflation regime (bias about +1.22), while SPF expectations fit that era extremely well (RMSE about 0.36), but the ranking reverses after 2020, when SPF underpredicts sharply (bias about −1.62) and the Michigan-based tightness-plus-energy specification tracks much more closely (RMSE about 0.56). The key insight is that "expected inflation" is not an interchangeable regressor: different proxies appear to capture different objects across regimes. In an anchored environment like 2008–2019, household expectations can embed persistent upward bias or slow-moving beliefs that do not map cleanly into price-setting. Professional forecasts, by contrast, track the anchored regime well but fail to adjust quickly enough to the post-2020 inflation process.

Panel B then clarifies the mechanism by switching the target to core PCE. The weaker role of the energy control for core is partly mechanical, energy is excluded by construction. It is still informative that the

import-price proxy delivers only modest gains for core in this short post-2020 window. A natural interpretation is that the large post-pandemic cost-push forces operated primarily through energy and other relative-price movements that load heavily on headline inflation, whereas core dynamics are captured mainly by expectations and labour-market tightness.

5.3.1 Fiscal Extension(Specification Robustness)

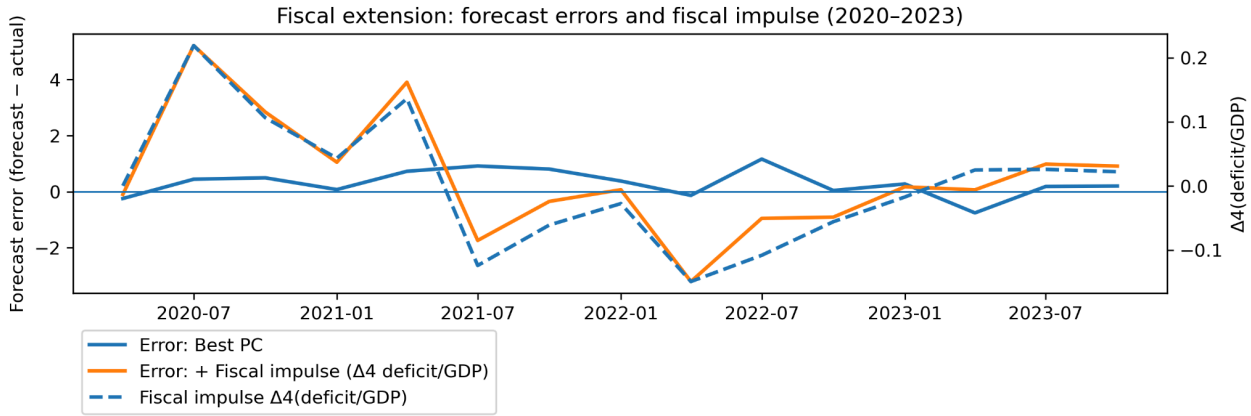

Figure 5: Forecast errors plot - showing best performing Phillips Curve, Phillips Curve with fiscal impulse control and an overlay of the fiscal impulse

Exhibit 5 plots forecast errors from the preferred Phillips-curve specification and from the same model augmented with a fiscal impulse $\Delta_4((G - T)/GDP)$, alongside the fiscal impulse series. Adding the fiscal proxy substantially worsens out-of-sample accuracy (RMSE rises from 0.56 to 2.13), and the fiscal-model errors closely co-move with the fiscal impulse itself, indicating that the fiscal regressor is driving forecast variation rather than absorbing residual inflation dynamics. In the pre-2020 estimation, the fiscal impulse loads with a large coefficient [δ= 22.84, s.e.= 11.845, see Appendix Table A1], consistent with the model extrapolating strongly from an endogenous stance variable. This extension therefore points to an identification limitation of reduced-form fiscal proxies in a short post-pandemic window: realised deficits respond to the cycle, and inflation mechanically affects measured deficits through nominal receipts and spending. Finally, this exercise does not test FTPL directly, which concerns fiscal news and beliefs about long-run adjustment and the monetary-fiscal regime; it is included as a compact diagnostic given the prominence of fiscal explanations in the post-pandemic literature and the scale of stimulus during 2020–21.

6. Conclusion

This note tested whether the Phillips curve “failed” to predict U.S. inflation after the pandemic, or whether standard implementations missed key post-2020 forces. The evidence tends to favour the latter interpretation: baseline Phillips-curve forecasts underpredict the 2021-22 surge, but performance improves materially once expectations and post-pandemic cost-push and slack measurement issues are addressed.

The first diagnostic concerns expected inflation. Holding the slack fixed, post-2020 tracking is highly sensitive to the expectations proxy: Michigan household expectations outperform professional and market-based measures in the pandemic window, while the Michigan mean-median wedge highlights that distributional shifts in household beliefs matter for baseline specifications.

Second, literal "failure" is assessed by benchmarking against inflation-only models. Simple persistence forecasts (no-change, trailing means, AR(1)) perform surprisingly well through parts of 2020-23, and the baseline unemployment-gap Phillips curve does not reliably dominate them, with conclusions varying by evaluation window. This aligns with the forecasting literature's message that Phillips-curve regressors often add limited incremental predictive content beyond inflation persistence, and motivates treating the post-2020 miss as a specification problem rather than a rejection of the mechanism.

Augmentations targeted at that problem deliver the clearest improvements. First, adding cost-push proxies substantially reduces headline forecast errors: Hazell-style energy inflation controls and bottlenecks-style import-price inflation both improve tracking relative to the baseline, consistent with large relative-price and supply forces shifting the inflation process after 2020. Second, slack measurement matters: replacing the unemployment gap with vacancy-based tightness improves performance in the pandemic window, and combining tightness with supply proxies yields the best headline results and smaller bias. Robustness checks reinforce that "failure" is not regime-invariant: the expectations proxy that performs best in 2020-23 need not generalise to 2008-2019, and the "supply rescue" is largely a headline phenomenon, with much weaker gains for core PCE.

The analysis is deliberately reduced-form and forecasting-oriented: supply proxies and tightness measures improve tracking but are not identified structural shocks, and the post-2020 window is short and dominated by one extraordinary episode, limiting power for slope-break and nonlinearity tests. Expectations and slack are proxied rather than observed, and the regime flips across Michigan and SPF expectations point to changing expectation formation or policy regimes. Future work could address these limitations by using identified shocks and higher-frequency evidence to separate demand, supply, and policy channels; particularly for fiscal explanations. For example, event-study or high-frequency designs that construct fiscal news shocks (as in Hazell and Hobler's approach) provide a cleaner test of FTPL-style mechanisms than contemporaneous deficit impulses. More generally, combining this pseudo out-of-sample framework with identified cost-push shocks and a model of expectation formation would clarify when Phillips-curve forecasts fail because the mechanism breaks versus when the proxies and omitted shocks change.

Word Count: 4998 words (2498 excluding 5 figures/tables)

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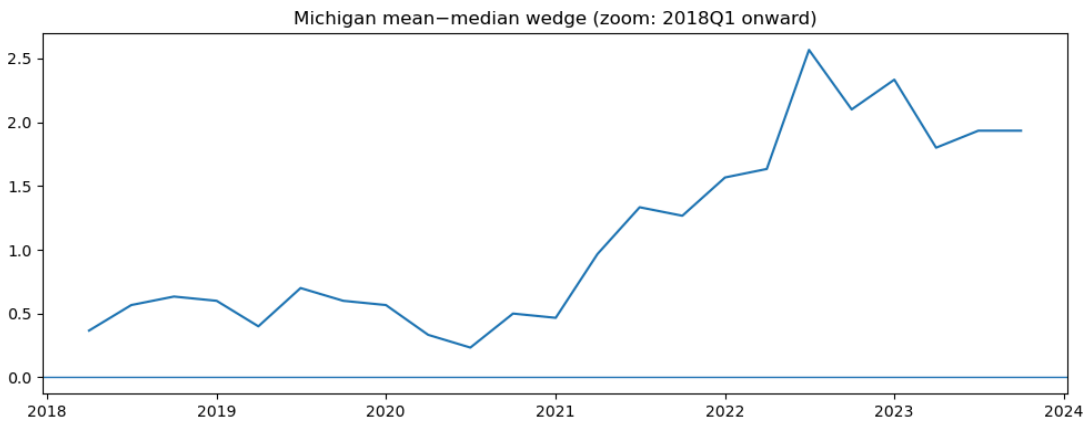
- I used OpenAI's ChatGPT (GPT-5.1 Thinking) to help debug Python code, refine phrasing in the write-up, and brainstorm the structure of the empirical strategy; all empirical implementation, result selection and interpretation are my own.

Appendix

Figure 1: Michigan mean-median wedge



Appendix Figure 1: Michigan expectations mean-median wedge post 2018Q1 (Exceptionally positively-skewed during the pandemic period)

Figure 2/Table A1: Fiscal extension coefficients

Appendix: Fiscal extension coefficient output

Baseline (train ≤2019Q4):

Variable	Coef	SE	t
exp_mich_1y_mean	1.023	(0.082)	12.49
log_vu	-0.082	(0.316)	-0.26
pi_pce_energy_yoy	0.041	(0.010)	3.89

+ Fiscal impulse Δ4(deficit/GDP) (train ≤2019Q4):

Variable	Coef	SE	t
exp_mich_1y_mean	1.021	(0.078)	13.03
log_vu	0.159	(0.256)	0.62
pi_pce_energy_yoy	0.043	(0.011)	3.81
fiscal_impulse_d4	22.838	(11.845)	1.93